

SentinelWire™ Operator Manual – End-to-End Workflow

Purpose & Scope

This Operator Manual provides a complete, end-to-end operational guide for using SentinelWire™ as a SOC-grade network investigation framework. It documents the workflow from environment personalization to IOC extraction, scoring, MITRE ATT&CK; mapping, and analytical pivoting.

Architecture Overview

SentinelWire™ combines personalized Wireshark profiles with automated PowerShell-based IOC extraction. The framework is designed for rapid triage, repeatability, analyst consistency, and explainable detections.

Step 1 – Environment Personalization (make_ws_profile.bat)

- Run make_ws_profile.bat on the target analyst workstation.
- Provide: Analyst IP, Router IP, Analyst MAC, Router MAC.
- Purpose: Enable router-aware and host-aware display filters.
- Outcome: Profiles optimized for 'to-me', 'from-me', and MITM detection.

Step 2 – Initial Capture (UNIVERSAL-DEFENSE)

- Select the active network interface in Wireshark.
- Apply the UNIVERSAL-DEFENSE profile.
- Capture live traffic for 30–60 seconds only.
- Stop capture immediately to preserve analyst focus.

Step 3 – IOC Extraction (extract_iocs_UPDATED_v1.1.ps1)

- Execute the PowerShell script against the captured PCAP.
- A unique run folder is created automatically.
- Artifacts extracted: DNS queries, TLS SNI, HTTP hosts, User-Agents, IPs, JA3 (if available).
- Candidate base64 strings are heuristically identified.

Step 4 – Automatic IOC Confidence Scoring

Each extracted artifact is scored (0–100) using a weighted model combining persistence, cross-protocol correlation, IOC type risk, and known-good reduction. Scores allow analysts to prioritize attention without manual enrichment.

Score Interpretation Guide

- 0–29: Low confidence (likely benign or noise).
- 30–69: Medium confidence (suspicious, contextual validation required).
- 70–100: High confidence (probable malicious activity).

Step 5 – MITRE ATT&CK; Technique Mapping

High- and medium-confidence IOCs are automatically mapped to MITRE ATT&CK; techniques based on observed network behavior rather than static indicators.

Common Technique Examples

- DNS tunneling → T1071.004
- ARP poisoning / MITM → T1557.002, T1040
- Network scanning → T1046
- Encrypted command-and-control → T1573, T1071.001

Step 6 – Analyst Pivoting Strategy

Based on dominant techniques and IOC clusters, analysts pivot from UNIVERSAL-DEFENSE to specialized Wireshark profiles to deepen inspection and validate hypotheses.

Profile Pivot Matrix

- MITM indicators → MITM-MASTER / ARP-SPOOF
- DNS anomalies → DNS-SUSPECT / LONG-DNS
- Reconnaissance patterns → SYN-WEIRD / TO-ME-NONROUTER
- Encryption abuse → TLS-ANOMALY

Operational Best Practices

- Keep captures short to reduce cognitive overload.

- Trust scoring for prioritization, not final verdicts.
- Document pivots and rationale for incident reports.
- Reuse the workflow consistently across investigations.

Analyst Output & Reporting

SentinelWire™ outputs (CSV, TXT summaries, MITRE mappings) are designed to be directly embedded into SOC tickets, DFIR reports, and executive summaries without reformatting.

Conclusion

SentinelWire™ enables analysts to move from packet noise to structured adversary behavior analysis in minutes, while preserving transparency, repeatability, and analytical rigor.